

Benchmarking without Time Travel: Public-Exposure Auditing and Temporal Generalization in Molecular AI

Author One,^{1,*} Author Two¹ and MolTrustBench Consortium¹

¹Department, Institution, Street, Postcode, State, Country

* Corresponding author. <mailto:author@example.com>

Abstract

Molecular AI benchmarks are commonly interpreted as prospective generalization tests, but held-out molecules, scaffolds, close chemical neighbors, and assay-derived labels can already be publicly observable before a relevant benchmark cutoff. This creates a temporal-validity problem that is distinct from conventional ADMET, out-of-distribution, class-imbalance, or activity-cliff benchmarking. We introduce MolTrustBench, a reproducible audit protocol and reporting standard that builds ChEMBL release-time indexes, annotates exact, scaffold, and nearest-neighbor public exposure, constructs exposure-aware evaluation slices, audits assay provenance, and emits benchmark trust cards. In a first ChEMBL24/27/30/33/36 audit of MoleculeNet BBBP, ClinTox, BACE, and Tox21, public exposure is measurable at multiple structural levels and varies strongly by task. Exposure-aware evaluation shows task- and model-family-dependent score sensitivity, while density, temporal, and train-label-shuffle controls define interpretation boundaries. ChEMBL36 assay provenance reveals duplicate records, unit heterogeneity, threshold sensitivity, and provenance-level label discordance for benchmark-overlapping and endpoint-native toxicology tasks. MolTrustBench reports observable public-exposure lower bounds, not model-specific training inclusion or memorization, and provides machine-readable supplement tables, data-freeze manifests, and trust-card templates for benchmark reporting.

Keywords molecular AI, benchmark trustworthiness, public exposure, temporal validity, ChEMBL, assay provenance

Abbreviations ADMET: absorption, distribution, metabolism, excretion, and toxicity; NN: nearest neighbor; TDC: Therapeutics Data Commons; AUROC: area under the receiver operating characteristic curve

Key Points

- MolTrustBench audits benchmark temporal validity rather than proposing another ADMET leaderboard.
- ChEMBL release-time indexes provide observable lower bounds for molecule, scaffold, and nearest-neighbor public exposure.
- Exposure-adjusted evaluation and assay-provenance diagnostics reveal benchmark trustworthiness risks that are invisible in single leaderboard scores.

Introduction

Molecular AI benchmarks increasingly serve as proxies for prospective drug-discovery and toxicology generalization. This interpretation assumes that held-out molecules represent chemistry that was not publicly observable at the relevant evaluation boundary. In the foundation-model era, however, a molecule can be absent from a supervised training split while still being publicly observable through ChEMBL, PubChem, ZINC, benchmark repositories, or assay-derived records before evaluation [3, 8, 7]. This distinction matters because a leaderboard score can mix prospective generalization with interpolation over publicly visible chemistry.

Existing molecular benchmark work has made substantial progress on data scarcity, out-of-distribution generalization, scaffold splits, class imbalance, activity cliffs, and ADMET reliability [14, 6, 2]. Recent molecular foundation-model work has also treated benchmark

overlap with public molecular corpora as a concrete evaluation-control issue [9]. MolTrustBench addresses a different validity question: whether the benchmark itself should be trusted as a temporal evaluation of chemistry not observable in an indexed public release history before a cutoff. We therefore use conservative terminology throughout. A ChEMBL release date is an observability date, not a model-specific pretraining date; public exposure is an observable lower bound, not proof that a particular model used or recalled a molecule.

Interpretation limits.

Public observability does not establish model training inclusion, memorization, or inappropriate benchmark use. MolTrustBench therefore audits an observable public-exposure lower bound and reports exposure-adjusted score sensitivity. These outputs should

Table 1 Novelty boundary for MolTrustBench. The paper is framed as a benchmark-trustworthiness protocol and reporting standard, not as a replacement for existing molecular performance benchmarks.

Adjacent benchmark area	MolTrustBench boundary
ADMET reliability benchmarks	Audits temporal observability and assay provenance rather than ranking ADMET predictors under scarcity, imbalance, or endpoint difficulty.
Molecular OOD and scaffold-split benchmarks	Reports whether test chemistry was publicly observable before a cutoff; this is complementary to distribution-shift or scaffold generalization tests.
Activity-cliff benchmarks	Treats scaffold and nearest-neighbor exposure as public-observability views, not as a substitute for local structure–activity difficulty.
Contamination or memorization studies	Provides observable exposure lower bounds and exposure-adjusted evaluations; it does not prove training-set inclusion or memorization.
Foundation-model evaluations	Uses frozen embedding wrappers only as protocol applicability checks, not as evidence that a checkpoint used ChEMBL.
Drug-discovery agent benchmarks	Does not evaluate autonomous discovery agents; it audits whether benchmark evidence is temporally and provenance-aware.

MolTrustBench workflow: from public observability to exposure-adjusted reporting

ChEMBL release dates are treated as public observability dates, not model-specific pretraining dates.

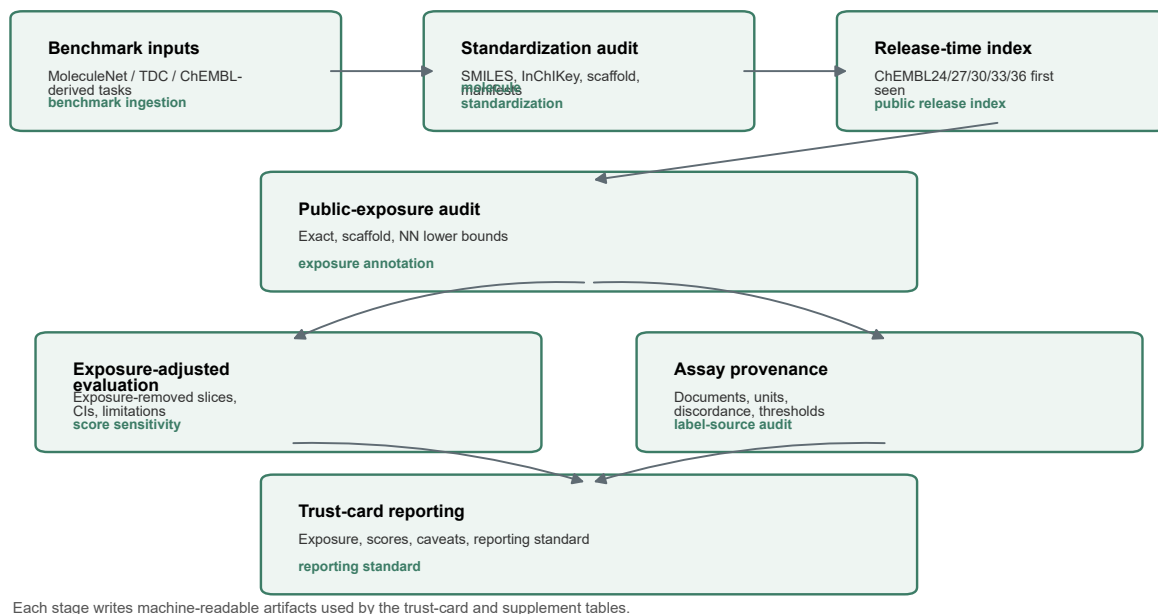


Figure 1 MolTrustBench workflow. MolTrustBench converts benchmark molecules and labels into auditable public-observability, exposure-adjusted evaluation, assay-provenance, and trust-card artifacts. Benchmark molecules are standardized, linked to ChEMBL release-time compound and scaffold indexes, annotated for exact, scaffold, and nearest-neighbor public exposure, evaluated under exposure-aware splits, and summarized in benchmark trust cards. ChEMBL release dates are treated as public observability dates, not as evidence that a specific model used ChEMBL during pretraining.

be interpreted as benchmark-trustworthiness diagnostics, not as accusations about a particular model or checkpoint.

MolTrustBench contributes a reproducible audit framework for benchmark trustworthiness. The framework builds historical release indexes, annotates exact molecule, scaffold, and nearest-neighbor exposure, constructs exposure-aware and temporal-future evaluation slices, audits assay provenance from ChEMBL SQLite, and generates benchmark trust cards. This first MolTrustBench audit uses real ChEMBL release indexes, four MoleculeNet case studies, endpoint-native toxicology pilots, negative controls, and machine-readable trust-card outputs to illustrate the protocol rather than to introduce a new ADMET leaderboard.

Methods

Release-time public-exposure index

We use historical ChEMBL releases as public-observability anchors. For each release, MolTrustBench records standard InChIKey, standardized SMILES, release metadata, and Bemis–Murcko scaffold information [3, 1]. Compound-level and scaffold-level indexes are collapsed to first-seen release and first-seen date. The initial index covers ChEMBL24, ChEMBL27, ChEMBL30, ChEMBL33, and ChEMBL36.

Benchmark standardization and exposure annotation

Benchmark molecules are standardized with RDKit when available [11]. Each molecule is annotated with exact public exposure, scaffold public exposure, nearest-neighbor exposure at Tanimoto thresholds 0.6, 0.7, 0.8, and 0.9, maximum nearest-neighbor similarity before the cutoff, earliest exact release, earliest scaffold release, exposure category, and exposure risk score. The main cutoff in this audit is ChEMBL30, corresponding to February 2022 in the ChEMBL release table.

Exposure-adjusted evaluation

For the current performance evidence, MolTrustBench compares standard splits against exact-removed, scaffold-removed, NN-removed, density-matched exposure-removed, and observable temporal-future subsets. Classical Morgan-fingerprint baselines use logistic regression and an XGBoost-style gradient-boosted classifier fallback [12, 4, 10], while sequence-neural baselines include SMILES-CNN, SMILES-GRU, and SMILES-Transformer families. Regression runs report RMSE, MAE, and R^2 under the same exposure-aware slice logic. These models are used to test score sensitivity under exposure-aware evaluation; they are not treated as evidence of foundation-model training exposure.

Train-label-shuffle controls permute training labels before fitting sequence baselines, then compare null-control exposure-adjusted deltas with observed-label deltas. This analysis is used as a negative control for score sensitivity, not as evidence that public exposure caused a specific model behavior.

For endpoint-native temporal evaluation, MolTrustBench also constructs ChEMBL-derived hERG, CYP3A4, CYP2D6, CYP2C9, CYP1A2, CYP2C19, and DILI/hepatotoxicity tasks from release-indexed bioactivity or text-evidence records. These tasks use the same exposure-aware split logic but are interpreted as endpoint-specific temporal-validity and provenance checks. If a strict exposure-removed slice becomes too small or one-class, the run is retained as an expected limitation rather than being promoted to a performance result. Frozen foundation-model embeddings are treated as checkpoint-light wrapper case studies; they are not used as evidence that a specific checkpoint used ChEMBL during pretraining.

Assay provenance

Assay-provenance auditing uses ChEMBL36 SQLite activity tables to recover assay identifiers, document identifiers, targets, assay types, standard measurement fields, units, relations, pChEMBL values when present, confidence scores, and source release. MolTrustBench summarizes duplicate compound counts, conflicting derived label calls, unit inconsistency, threshold sensitivity, document-level heterogeneity, and a derived-label discordance score. The current evidence includes MoleculeNet-overlapping BBBP and ClinTox molecules plus endpoint-native ChEMBL-derived hERG, CYP1A2, CYP2C19, and DILI/hepatotoxicity endpoints.

Results

MolTrustBench establishes a release-time index for public-exposure auditing

MolTrustBench first constructs a release-time index that converts public chemical database history into an auditable benchmark

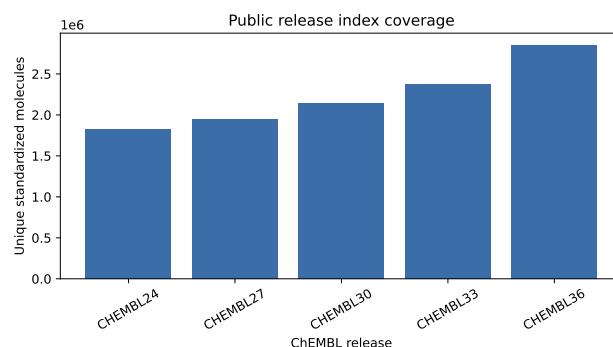


Figure 2 ChEMBL release-time index. ChEMBL24/27/30/33/36 define a reproducible timeline for observable public exposure. The release index records first-seen compound and scaffold observability after deterministic molecule standardization, enabling benchmark test molecules to be audited against a fixed release cutoff. These dates support public-exposure lower bounds rather than model-specific pretraining cutoffs.

Table 2 ChEMBL release index used for the first MolTrustBench audit.

Release	Date	DOI suffix	Input	Std. molecules
CHEMBL24	2018-05	database.24	chemreps	1,820,035
CHEMBL27	2020-05	database.27	chemreps	1,941,410
CHEMBL30	2022-02	database.30	chemreps	2,136,187
CHEMBL33	2023-05	database.33	chemreps	2,372,674
CHEMBL36	2025-07	database.36	chemreps	2,854,815

timeline. We indexed five ChEMBL releases spanning ChEMBL24, ChEMBL27, ChEMBL30, ChEMBL33, and ChEMBL36, with release dates from May 2018 to July 2025 and standardized molecule counts ranging from 1.82 million to 2.85 million records (Table 2). For each release, the pipeline records standard InChIKey, standardized SMILES, release metadata, and Bemis-Murcko scaffold information, then collapses records into earliest observable release indexes at the compound and scaffold levels. This design intentionally treats ChEMBL release date as an observability date, not as proof that any specific model used ChEMBL during pretraining.

Public exposure is measurable across benchmark tasks

The real-data audit finds measurable public exposure across all four completed MoleculeNet tasks, supporting the claim that benchmark trustworthiness should be audited before interpreting held-out molecules as prospectively unseen chemistry. At the ChEMBL30 cutoff, exact exposure was 0.838 for BBBP, 0.474 for ClinTox, 0.297 for BACE, and 0.921 for Tox21 (Table 3). The four-task audit covers 12,290 benchmark molecules, with ChEMBL-index-unobserved exact counts ranging from 331 in BBBP to 1,063 in BACE. These values are not uniform across tasks: BACE contains a larger exact-unobserved subset, whereas Tox21 is already highly exact-exposed under the same cutoff. This heterogeneity is the point of the audit: a benchmark trust card should report the public-exposure profile of each task rather than assuming that all benchmark test sets carry the same temporal meaning.

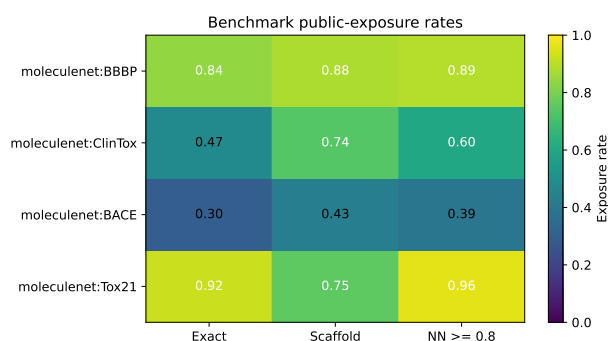


Figure 3 Multi-level public-exposure landscape. Exact, scaffold, and nearest-neighbor exposure rates vary across MoleculeNet BBBP, ClinTox, BACE, and Tox21 at the ChEMBL30 cutoff. The heatmap shows that exact molecule occurrence is only one exposure view: scaffold and NN exposure can reveal additional observable exposure lower bounds, while some tasks such as Tox21 are already highly exact-exposed. The result motivates per-benchmark trust cards instead of a single universal exposure label. Denominators are the audited benchmark molecule counts in Table 3.

Table 3 Audit-only exposure coverage at the ChEMBL30 cutoff.

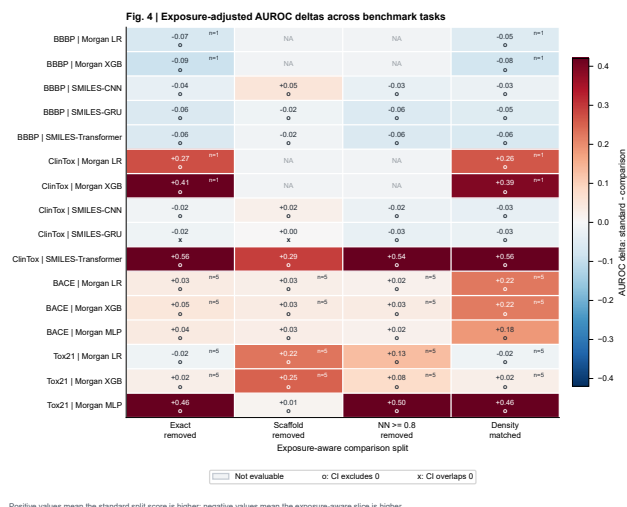
Task	Molecules	Exact	Scaffold	NN@0.8
BBBP	2,039	0.838	0.878	0.889
ClinTox	1,480	0.474	0.737	0.599
BACE	1,513	0.297	0.427	0.394
Tox21	7,258	0.921	0.753	0.959

Scaffold and nearest-neighbor exposure extend beyond exact matching

Exact molecule matching is only the most conservative exposure test, and the scaffold and nearest-neighbor views provide additional public-observability views. For BBBP, exact exposure at ChEMBL30 was 0.838, while scaffold exposure was 0.878 and nearest-neighbor exposure at Tanimoto ≥ 0.8 was 0.889. ClinTox showed a larger gap between exact and structural exposure: exact exposure was 0.474, but scaffold exposure reached 0.737 and NN@0.8 reached 0.599. BACE showed the same ordering at lower absolute rates, with exact exposure of 0.297, scaffold exposure of 0.427, and NN@0.8 exposure of 0.394. Tox21 is an important counterexample rather than an inconvenience: exact exposure was already very high at 0.921, while scaffold exposure was 0.753 and NN@0.8 was 0.959. These results indicate that public-exposure auditing is multi-level and task-specific, not a single monotonic score.

Exposure-adjusted evaluation changes benchmark interpretation

Exposure-adjusted evaluation shows that standard split scores and exposure-removed subset scores can diverge enough to change how benchmark performance is interpreted. On BBBP, Morgan-logistic regression achieved AUROC 0.879 on the standard split and 0.946 on the exact-removed split; Morgan-XGB similarly changed from 0.885 to 0.976. ClinTox moved in the opposite direction: Morgan-logistic regression changed from AUROC 0.836 on the standard split to 0.565 on the exact-removed split, and Morgan-XGB changed from 0.784 to 0.370. These results are not interpreted as a universal performance drop. Instead, exposure-aware subsets can produce different estimates



Positive values mean the standard split score is higher; negative values mean the exposure-aware slice is higher.

Figure 4 Exposure-adjusted score deltas. Exposure-adjusted evaluation changes performance interpretation across tasks, splits, and model families. Deltas are computed as standard-score minus comparison-score for matched standard and exposure-aware slices, with bootstrap or seed-level uncertainty where available (source tables S6, S7, and S29). The figure is evidence of benchmark score sensitivity under public-exposure auditing; it does not prove model-specific training exposure or label recall.

of model performance, with direction and magnitude depending on task composition, label structure, model class, and exposure-removed subset size.

Density-matched and temporal-future controls define the current evidence boundary

The first controls address two interpretation risks: public exposure may be a proxy for chemical-space density, and release dates are not model-specific training dates. Density-matched exposure-removed controls did not eliminate score differences in the evaluated tasks. For BACE, Morgan-XGB AUROC changed from 0.767 on the standard split to 0.720 on exact-removed and 0.550 on the density-matched split. For Tox21, Morgan-XGB changed from 0.707 on standard to 0.849 on exact-removed, 0.584 on scaffold-removed, and 0.849 on density-matched, while the exposure-removed subsets were much smaller than the standard test set. Temporal-future splits remain a narrower control: BBBP had 34 temporal-future test molecules, ClinTox had only 8, and BACE/Tox21 temporal-future subsets were too small or single-class for stable classification metrics. These controls support the framework logic, while temporal-generalization evidence remains a current boundary rather than a final deployment claim.

Endpoint-native ChEMBL temporal tasks strengthen the temporal-validity check

The endpoint-native ChEMBL temporal wave adds a more direct test of temporal validity because the tasks are derived from ChEMBL bioactivity records rather than mapped retrospectively from MoleculeNet labels. The integrated run contains SMILES-CNN, SMILES-GRU, and SMILES-Transformer metrics across hERG, CYP3A4, CYP2D6, and CYP2C9 endpoint-native tasks, with standard, exact-removed, scaffold-removed, NN@0.8-removed,

density-matched, and temporal-future slices where valid. Coverage is complete for observed task/model/split groups. Expected-limitation rows come from one-class exposure-removed subsets after strict filtering, especially CYP2C9, which is precisely the kind of boundary a benchmark trustworthiness audit should reveal rather than hide.

The task summaries show substantial endpoint-level temporal drops, interpreted as endpoint-specific temporal-validity evidence rather than a universal model claim. Mean standard AUROC is around 0.89 for hERG sequence models, while the mean temporal drop is 0.19 for SMILES-CNN and 0.28 for SMILES-GRU. CYP3A4 and CYP2D6 also show positive temporal drops across sequence models, whereas CYP2C9 retains only standard, scaffold-removed, and temporal-future groups because exact/NN/density exposure-removed slices become class-degenerate. This supports the performance-sensitivity and sample-size boundary arguments together: exposure-adjusted evaluation can reveal performance changes, and the same framework records when exposure-removed slice construction is not statistically usable.

Frozen FM embeddings define a foundation-model wrapper boundary

The checkpoint-light foundation-model case study tests whether the same exposure-aware contracts can be applied to staged molecular FM wrappers [13, 5]. A safetensors MolFormer workflow contributes 1,140 consolidated frozen-embedding metric rows with zero failure rows. A second safetensors wrapper, ChemBERTa-100M, contributes 150 frozen-embedding rows with zero failure rows across BACE, Tox21, BBBP, ClinTox, CYP1A2, and CYP2C19. This result documents wrapper applicability boundaries, but it remains a boundary claim: it does not show that MolFormer or ChemBERTa used ChEMBL during pretraining and does not generalize to all molecular foundation models. Earlier blocked FM attempts are recorded separately as artifact-format/runtime limitations, not scientific negative results.

Train-label-shuffle controls provide a negative-control check

Train-label-shuffle controls test whether exposure-adjusted deltas remain larger than a null training-label baseline. The overnight control run produced 360 valid null-control metric rows across Tox21 and the ChEMBL endpoint-native tasks, with SMILES-CNN, SMILES-GRU, and SMILES-Transformer baselines where applicable. It also produced 180 expected one-class limitation rows and zero critical failures. BACE is retained only as an expected one-class standard-split limitation under the submitted annotation configuration and is not interpreted as label-shuffle evidence.

The null-control comparison table summarizes 22 observed-vs-shuffle delta comparisons. Eighteen comparisons have larger absolute observed deltas than shuffled-label deltas, with a mean absolute observed delta of 0.163 versus a mean absolute shuffled-label delta of 0.038. Four comparisons remain null-sensitive, meaning the shuffled-label delta was not smaller than the observed-label delta. This mixed result shows that many exposure-adjusted deltas are larger than shuffled-label deltas while preventing the article from overstating a universal exposure effect. The paired null-control visualization is retained as supplementary figure source data rather than treated as a central main-text result.

A de-duplicated CYP label-shuffle integration provides paired null-control evidence for the CYP extension. The integrated

run contributes 360 train-label-shuffle metrics, 1,080 expected exposure-removed slice limitations, and zero critical failures across CYP1A2/CYP2C19 sequence baselines. The paired observed-vs-shuffled temporal-delta table contains 120 CYP2C19 rows across SMILES-CNN, SMILES-GRU, and SMILES-Transformer families. CYP1A2 did not yield paired temporal-delta rows in this snapshot and is reported as a coverage limitation.

Model-family sensitivity shows that exposure effects are not confined to one baseline

The model-family sweep extends exposure-adjusted evaluation beyond classical fingerprint models while keeping the interpretation conservative. Across BBBP and ClinTox, the current artifacts include Morgan logistic regression, Morgan-XGB, Morgan MLP, SMILES-CNN, SMILES-GRU, and SMILES-Transformer results under standard, exact-removed, scaffold-removed, NN-removed, density-matched, and temporal-future slices. On BBBP, SMILES-CNN AUROC changed from 0.914 on the standard split to 0.950 on exact-removed, 0.863 on scaffold-removed, and 0.947 on NN@0.8-removed. On ClinTox, SMILES-CNN AUROC changed from 0.905 on standard to 0.921 on exact-removed, 0.884 on scaffold-removed, and 0.930 on NN@0.8-removed. For BACE and Tox21, sequence-family integration covers SMILES-CNN, SMILES-GRU, and SMILES-Transformer architectures across repeated seeds and exposure-aware splits. Paired bootstrap and seed-level confidence intervals then separate stable exposure-adjusted deltas from uncertain deltas; for example, Tox21 SMILES-GRU exact-removed and density-matched deltas cross zero, while most BACE/Tox21 SMILES-CNN deltas do not. A regression extension adds ESOL standard-vs-exposure-aware RMSE deltas across the same sequence families; Lipophilicity is retained as an expected-limitation case because exposure-aware slices were too small or absent under the current cutoff. The goal of this analysis is not to rank model families; it is to show that benchmark trustworthiness conclusions should be reported by exposure slice and model family rather than by a single leaderboard number.

Assay provenance reveals label-reliability risk beyond molecule exposure

MolTrustBench also audits whether the molecules underlying benchmark labels have heterogeneous public assay records. ChEMBL36 SQLite extraction identified activity records for 1,289 BBBP molecules and 654 ClinTox molecules, with high duplicate and discordant derived-label counts among molecules with public activity records. For BBBP, the assay-provenance summary recorded 1,215 duplicated compounds, 677 molecules with discordant derived label calls, 932 molecules with unit inconsistency, and 728 threshold-sensitive molecules. For ClinTox, the corresponding counts were 648 duplicated compounds, 502 discordant derived label calls, 613 unit inconsistencies, and 524 threshold-sensitive molecules. The endpoint-native hERG pilot provides a cleaner assay-provenance case study: 4,999 ChEMBL-derived hERG molecules had activity coverage, with 1,591 duplicate compounds, 163 discordant derived label calls, and 2,646 threshold-sensitive molecules. CYP1A2 and CYP2C19 add two further endpoint-native ChEMBL-derived provenance summaries and temporal split manifests. A DILI/hepatotoxicity text-evidence endpoint adds 528 labeled activity records, 513 benchmark molecules, 514 molecule-level provenance summaries, one provenance-level binary text-label discordance before final label filtering, and 60

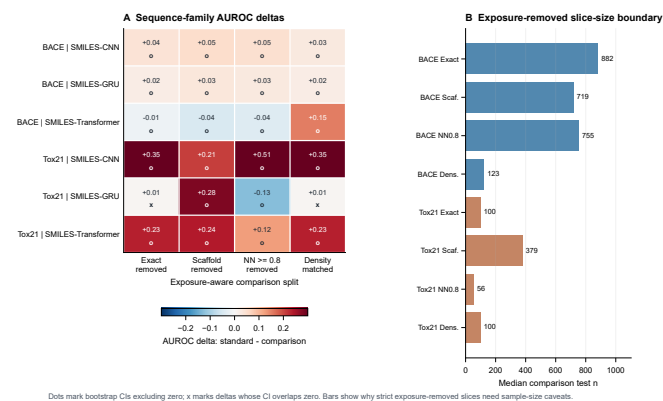


Figure 5 Sequence-family exposure sensitivity. BACE/Tox21 sequence-family baselines show that exposure-adjusted score changes are architecture- and task-dependent. In panel A, heatmap colors encode standard-minus-comparison AUROC deltas; open circles denote intervals excluding zero and x marks intervals crossing zero. In panel B, blue bars denote BACE comparison-slice sizes and orange bars denote Tox21 comparison-slice sizes. The integrated sweep covers SMILES-CNN, SMILES-GRU, and SMILES-Transformer baselines after run-identifier deduplication. These models are sequence baselines for exposure-adjusted evaluation, not foundation-model pretraining evidence.

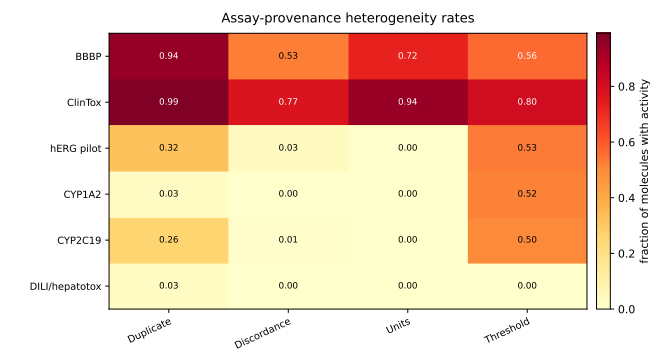


Figure 6 Assay-provenance heterogeneity map. ChEMBL36 SQLite assay provenance reveals duplicate activity records, unit heterogeneity, threshold sensitivity, and derived-label discordance for benchmark molecules with public assay records. BBBP and ClinTox rows represent public ChEMBL assay heterogeneity among overlapping molecules, while hERG, CYP1A2, CYP2C19, and DILI/hepatotoxicity provide endpoint-native ChEMBL-derived provenance cases. Denominators are molecules with matched public activity records, and flag definitions are given in Table S26. The figure supports label-source reliability auditing, not a claim that original benchmark labels are simply wrong.

standard/scaffold-removed sequence metrics with zero failure rows. Its exact-removed, NN@0.8-removed, temporal-future, and density-matched slices shrink to one-class two-molecule tests and are reported as expected exposure-removed slice limitations. Together, these results support the claim that benchmark trustworthiness is not only a molecule-exposure problem; it also depends on assay source, measurement units, censoring relations, thresholding choices, text-label evidence, and document provenance.

Benchmark trust-card examples
Trust cards expose public observability and exposure-removed subset limits before leaderboard scores are interpreted.

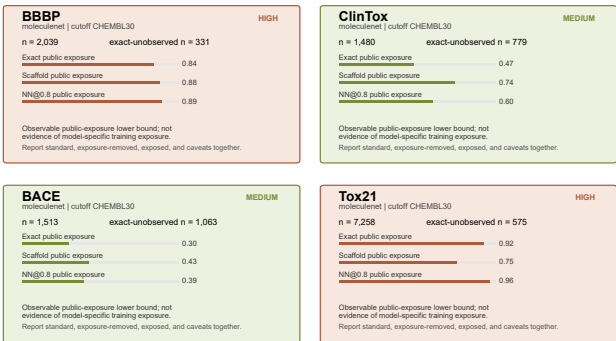


Figure 7 Benchmark trust-card examples. Benchmark trust cards summarize public-exposure rates, exposure-removed subset sizes, exposure-adjusted score evidence, assay-provenance availability, and audit caveats for individual benchmark tasks. The card format turns MolTrustBench outputs into a reporting standard so benchmark users can inspect temporal-validity and provenance limitations before interpreting leaderboard scores as prospective generalization. Machine-readable schema and JSON examples are provided in Table S28 and the supplement trust-card directory.

Table 4 Minimum reporting checklist.

Item	Required report field
Public cutoff	Release or date used as the public-observability boundary
Exposure rates	Exact, scaffold, and nearest-neighbor public exposure
Exposure-removed subset	Exact-removed, scaffold-removed, and NN-removed test sizes
Assay provenance	Assay IDs, document IDs, units, relations, and derived-label discordance scores when available
Model metadata	Claimed pretraining corpus and cutoff, or explicit unknown status
Adjusted scores	Standard, exposure-removed, exposed, temporal-future, and density-matched scores
Limitations	Whether evidence is audit-only, performance-tested, or endpoint-native

Trust cards convert the audit into a reusable reporting standard

The final result is a reporting layer that turns exposure annotations, exposure-adjusted metrics, assay provenance, and sanity checks into benchmark trust cards. For each audited task, MolTrustBench records exact exposure, scaffold exposure, NN exposure at multiple thresholds, exposure-removed subset sizes, cutoff release, assay-provenance availability, and audit caveats. The report card framing is important because this article is not proposing a new ADMET leaderboard. Instead, it asks benchmark users to report whether their test molecules and labels were publicly observable before a relevant cutoff and whether performance estimates are stable under exposure-aware slices.

Discussion

MolTrustBench reframes molecular benchmark evaluation as a trustworthiness audit. The central finding is not that any model used a specific molecule or label, but that commonly used benchmark tasks contain measurable public-exposure and assay-provenance risks that should be reported before leaderboard scores are interpreted as prospective generalization. This framing separates three questions that are often conflated: whether chemistry was publicly observable, whether model performance changes under exposure-aware evaluation, and whether assay-derived labels are stable enough to support benchmark claims.

The current evidence package is intentionally scoped. The four-task audit supports the molecule, scaffold, and nearest-neighbor public-observability claims. Exposure-adjusted evaluation is supported by the four core tasks, BACE/Tox21 sequence-family coverage, ESOL regression robustness, endpoint-native hERG and CYP temporal metrics, CYP1A2/CYP2C19 sequence baselines, frozen MolFormer and ChemBERTa-100M wrapper checks, and train-label-shuffle negative controls. Assay-provenance claims are supported by ChEMBL36 SQLite extraction plus endpoint-native hERG, CYP1A2, CYP2C19, and DILI toxicology evidence. The reusable reporting-standard contribution is supported by the workflow schematic, generated trust cards, plotted trust-card examples, sanity checks, data-freeze records, and machine-readable supplement tables S24–S29.

The main limitations are also explicit: SIDER remains deferred from the first coverage expansion; Lipophilicity exposure-aware regression slices are too small or absent under the current cutoff; CYP2C9 exact/NN/density exposure-removed slices become one-class subsets in the endpoint-native wave; DILI exact/NN/temporal/density exposure-removed slices are one-class two-molecule tests; BACE label-shuffle controls are class-degenerate under the submitted annotation configuration; four observed-vs-null comparisons remain null-sensitive; CYP1A2 does not form paired temporal-delta rows in the CYP label-shuffle integration; temporal-future subsets are small for some tasks; frozen FM rows are wrapper applicability evidence rather than model-specific pretraining evidence; and model-specific pretraining cutoffs remain unknown for many public checkpoints.

These limitations are audit outputs rather than failures to be hidden. Exposure-removed subsets that become too small or class-degenerate under strict filtering are evidence about benchmark usability under exposure-adjusted evaluation. MolTrustBench provides a reproducible scaffold for replacing implicit benchmark assumptions with explicit public-exposure lower bounds, assay-provenance diagnostics, and exposure-adjusted reporting.

Data availability

Processed evidence tables, exposure annotations, generated benchmark splits, figure source data, supplement tables, trust-card JSON examples, and analysis manifests underlying this article will be available from a DOI-issuing repository at [repository DOI/accession to be inserted]. The corresponding local evidence archive pending repository deposit is `moltrustbench_data_freeze_20260529_161700.tgz`; its SHA256 checksum is recorded in the deposit manifest `moltrustbench_data_freeze_20260529_161700_manifest.csv` and in the supplement data-availability file.

The supplement package includes compact source data, slice-validity tables, standardization audit tables, assay-provenance flag definitions, machine-readable trust-card schemas, and a figure source-data map. The archive excludes raw ChEMBL database dumps, local secrets, transient caches, and third-party model checkpoints. ChEMBL source releases are available from the official ChEMBL downloads page [3]; benchmark datasets and public checkpoints should be cited through their original sources.

Code availability

MolTrustBench code is maintained in the project repository. The current version uses the release-index, exposure-audit, assay-provenance, model-family integration, label-shuffle negative-control, supplement-packaging, and final-figure scripts under `scripts/`, with reusable package modules under `src/moltrustbench/`. Before submission, the GitHub commit SHA or tagged release corresponding to the deposited evidence archive should be added here.

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Conflict of interest

The authors declare no competing interests. This statement should be verified before submission.

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